

Evaluating in-silico perturbation with Geneformer

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Method. Geneformer encodes each cell as a list of its genes ranked by expression relative to each gene’s median across the pretraining corpus (zero-count genes omitted; at most 4,096 tokens for V2 models). We fine-tune it to separate a donor’s tumour-infiltrating T cells from that donor’s normal-tissue T cells, with donors held out. In-silico perturbation (ISP) deletes a gene from a tumour cell’s list or moves it to the top, and scores the change in cosine similarity of the cell’s embedding to the centroid of the same donor’s normal T cells. No cell is modified, and the model returns a score for every gene.

Six criteria, each traced to a documented error (lab standard ISP-STD-1, 28 September 2026)

Criterion	Rule	Evidence behind it
1. Register first	Test, direction, threshold, reachable p and outcome wording fixed and reviewed before any result	An August ordering was interpreted without a plan
2. Instrument works	Held-out balanced accuracy ≥ 0.60 with a donor sign test; a no-op edit gives exactly 0	July tumour and normal groups came from different studies (0 of 18 supplied both)
3. Same cells	Deletion and overexpression analysed on identical cells (since 25 September)	Overexpression on 2,424 cells for every gene, deletion on 202 to 1,131
4. Matched controls	Each gene against at least 20 genes of similar detection and rank; ambient-RNA genes flagged	S100A8/S100A9 shifted cells the same way under both edits in 9 of 12 comparisons
5. Random genes	Opposed effects of the two edits are not, on their own, evidence	Random genes: Spearman rho -0.593 (100 genes), -0.608 (200)
6. Donors, stability, fixed names	Donor as unit; stable in 95% of 10,000 bootstrap draws; never “no effect”	One donor held 74.9% of the SCLC test cells

Supported. Lung (43 donors): held-out balanced accuracy 0.825, all donors above chance; 13 of 34 testable T-cell genes exceed matched controls, stably; opposed effects are generic. Colon (E2; Pelka et al. 2021, 19 donors with unsorted cells in both tissues, from 25 eligible of 62): registration passed, classifier gate passed (0.904; 19 of 19 donors; lowest 0.655), no-op shift 0.0. **Not shown:** that any gene regulates T-cell state; what the classifier reads (T-cell state, ambient RNA or subset composition); transfer of the lung models to new lung donors (E1 blocked: data not accessible).

Colon results (after review). Random genes: $p = -0.245$ ($p = 0.0067$), same direction as lung but stable in only 78% of resamples (bar 95%): `control_draw_sensitive_open`, open. Lung reference genes: 3 of 10 kept their direction (bar 9): `pattern_not_replicated`. Not attributable to tissue alone.

Bulk RNA-seq (two tests). Geneformer cannot take a bulk profile, so we tested a CellOracle-style linear network (transcription-factor-to-target edges from a promoter base network, filtered by matched ATAC-seq) against measured CRISPR knockouts in CD4+ T cells, twice, with the same registered rules. Freimer et al. 2022 (17 knockouts): median rho 0.083 ($p = 0.012$), 4 of 17 beat random transcription factors, but not a shuffled network: `RECOVERED_NONSPECIFIC`, matched by co-expression alone. Weinstock et al. 2024 (38 new knockouts, same laboratory): no test passed (median rho 0.033, $p = 0.21$): `NOT_RECOVERED`, so the first reading does not replicate. In both sets, networks fitted on unperturbed controls showed no agreement above chance and predicted shifts were 55 to 85 times too small. The sets differ in harvest timing, deduplication, counting pipeline and possibly donors, uncontrolled; bulk data cannot separate composition from regulation. The transposable-element arm was `NOT_TESTABLE` in that run (no deposited TE count table); see below for the re-alignment.

Transposable elements (Geneformer TE). A second project asks which regulators set the expression of transposable elements (TEs), the repeated sequences that cells keep silenced. The single-cell Geneformer track (TE states read out through a classifier and a probe, since Geneformer has no TE tokens) is built but has not run: no human single-cell data with TE counts have been prepared. The bulk network track was run on 162 libraries of a reef fish (*A. polyacanthus*; 3 tissues $\times$ 3 CO2 groups). Without controls it reported 32,142 regulator-to-family effects at $FDR < 0.05$; against 90 expression-matched random genes, none of 30 TE regulators passed (1 at empirical $p \leq 0.05$ where 1.5 are expected; Mann–Whitney $p = 0.70$), and one global TE axis (29.3% of TE variance) set the predicted directions. In an exploratory analysis, developmental CO2 lowered the gill TE share of reads from 2.87% to 2.55% (–11.4% [–17.2, –5.1], $FDR 0.006$); brain and liver showed no robust shift. **Zebrafish ground truth** (17 published contrasts, 14 GEO series, TEs re-counted, each study against its own controls): 0 of 16 loss-of-function studies shifted the global TE share at $FDR < 0.05$; the one shift, human UHRF1 overexpression in liver (+29.9% [16.6, 44.5], $FDR 0.011$), is mostly compositional (gene-assigned reads fell, TE reads did not change). In exploratory family-level tests, loss-of-function studies leaned toward derepression in 6 of 8 ($p = 0.29$), and 6 of 7 for DNA-methylation regulators ($p = 0.13$); none passes FDR , and three studies have genotype-linked mapping differences (post hoc flag). The fish network directions agreed with the zebrafish leans at chance level (4 of 9; Spearman $p = -0.18$, $p = 0.50$). **Freimer TE arm** (96 libraries re-aligned; gates passed): `TE_RECOVERED_NONSPECIFIC` ($P1 p 0.18$, $p = 0.001$; no better than a shuffled network, $p = 0.34$). A post hoc control-only null reaches most knockouts’ responsive-subfamily counts; only CBFB exceeds it ($p = 0.010$), against 14 of 17 knockouts for genes. 83% of TE counts are intronic, so TE shifts cannot be separated from host-gene transcription.

Still to test. Goal swap; subset re-analysis; an interpretable baseline from T-cell programs (TCAT, Kotliar et al. 2025); model size (104M vs 316M); positive controls and experimental screens; bulk prediction with a composition covariate or single-cell data; TE data from full-length total RNA; better-replicated zebrafish loss-of-function studies; the single-cell TE track. Possible methods paper, working title “Opposite by default”. Every number is sourced in `SOURCES.md`.